ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai - 119

L A B R E C O R D

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name: B Idhayachandran

Roll No: 26CS1264

Date:

Subject Name: Professional English Lab Integrated

Experiment No: I-1

Self Introduction: Written Script and One-Page Profile

B Idhayachandran

Aspiring Cybersecurity Analyst and Data Scientist

Links	GitHub: idhayachandran | LinkedIn: Idhayachandran
Summary	I’m curious about how systems break and how data can explain why. When a problem gets in my way, I read, test and adapt until it is solved. I’m interested in two connected fields: cybersecurity and data science. Security work produces huge volumes of logs and alerts, and data science is the best way to make sense of them. Both fields reward patience, evidence and clear thinking, and both are hard to automate away because they depend on judgement.
Introduction	Hi, I’m B Idhayachandran, a Computer Science Engineering student (batch 2026–2030) at St. Joseph’s College of Engineering, Chennai. I enjoy programming in Python and C, and I like to think a problem all the way through before I act. Down the line, I want to work in cybersecurity, using data science to detect threats and protect people’s information.
Skills	Python, C, problem solving, learning cybersecurity and data science fundamentals
Experience	[Add any project, hackathon or competition you have taken part in]
Education	B.E. CSE, 1st Year, St. Joseph’s College of Engineering, Chennai
Awards	[Add any award or certificate you have received]

Experiment No: I-2

Impressive Personality: Biographical Note

Elon Musk, entrepreneur and engineer; CEO of Tesla and SpaceX

Elon Reeve Musk was born on 28 June 1971 in Pretoria, South Africa. He is an entrepreneur and engineer who has founded or led companies in electric vehicles, space transport, satellite internet, brain-computer interfaces, tunnelling and artificial intelligence. He is the CEO of Tesla and SpaceX, and he is among the best-known business figures in the world.

Early life and education

Musk showed an early interest in computers. As a boy he taught himself programming and, at about the age of twelve, sold the source code of a simple space-themed video game called Blastar to a magazine. He left South Africa at seventeen and moved to Canada, where he studied at Queen’s University in Ontario. He later transferred to the University of Pennsylvania in the United States and graduated with degrees in economics and physics. He began a PhD programme in physics at Stanford University but left after a very short time to start a business.

Early companies

In 1995 Musk and his brother Kimbal founded Zip2, a company that supplied online business directories and maps to newspapers. Compaq bought it in 1999. Musk then co-founded X.com, an online bank, which merged with a rival to become PayPal. PayPal had to fight online payment fraud from its earliest days, so security and data analysis were central to its business. eBay bought PayPal in 2002.

SpaceX

In 2002 Musk founded Space Exploration Technologies, or SpaceX, with the aim of making space travel cheaper and eventually helping people live on other planets. After early failures, its Falcon 1 rocket reached orbit in 2008. In 2012 the company’s Dragon capsule became the first commercial spacecraft to dock with the International Space Station. SpaceX later developed reusable Falcon 9 boosters that return to Earth and land, which sharply cut the cost of launches. In 2020 its Crew Dragon spacecraft carried NASA astronauts to the Station. Its Starlink network provides satellite internet, and the company is also developing Starship, a very large reusable launch vehicle.

Tesla

Musk joined Tesla Motors in 2004 as an early investor and chairman, and he became its CEO in 2008. Tesla showed that electric cars could be fast, desirable and practical. Its Model S, Model 3 and Model Y helped push the whole car industry towards electrification. Tesla also makes battery storage systems and solar products, and it collects large amounts of driving data to train its driver-assistance software.

Other ventures

Musk co-founded Neuralink in 2016 to develop brain-computer interfaces and founded The Boring Company the same year to build cheaper tunnels. In 2022 he bought the social network Twitter, which was renamed X in 2023. In 2023 he launched xAI, an artificial intelligence company that builds the Grok chatbot. He also helped found OpenAI in 2015 but left its board a few years later.

Working style and outlook

Musk is known for working on very large problems and for reasoning from first principles, which means breaking a problem down to basic facts instead of copying what others do. He often sets very ambitious deadlines for his teams. He is also a controversial public figure, and his business decisions and public statements attract both strong praise and strong criticism.

Why he impresses me

As a student interested in cybersecurity and data science, I find several lessons in his career. He taught himself to code when he was young. His first big business success came in payments, where fraud detection and secure systems matter every day. His companies depend on data, from rocket telemetry to millions of vehicles’ driving records, and they all depend on software that must be secure and reliable.

Experiment No: II-1

Summary of Lecture Notes

NPTEL: Enhancing Soft Skills and Personality (Prof. T. Ravichandran, IIT Kanpur)

Week 1: Course Introduction and Growth Mindset

The course opens with two preview lectures that recap an earlier course, “Developing Soft Skills”. That course dealt with perception, self-management, excellence, conflict, stress, habits, listening, telephone skills, technology, communication barriers, non-verbal communication, presentations, reading and trust. The big new topic here is mindset: a person’s habitual attitudes and ways of reading a situation.

In the shoe-salesman story, nobody wears shoes. Is that “no chance” or a “splendid opportunity”? The same situation draws opposite responses. Drawing on Carol Dweck, the lectures set a fixed mindset against a growth mindset. People with a fixed mindset avoid challenges, resist change, dwell on their limits, reject criticism and stop learning once school is over. Growth-mindset people welcome challenges, learn from failure and treat learning as lifelong.

Learning mindsets form early. A belief like “I’m not smart enough” limits a person, but it isn’t a fact about their ability. The suggested “secrets” are to behave like a baby (open and unafraid), put up with critical people, reject negative stereotypes and stereotype threat, and look for a sense of belonging. The Hawthorne effect is offered as evidence that belonging and attention motivate people more than incentives do.

Week 2: Time Management, Procrastination and Assertiveness

Time can’t really be managed. What we manage is our habits and our perception of it. Monochronic people stick to rigid schedules, while polychronic people stay flexible. Others tend to judge our personality by our punctuality and by how we behave under deadline pressure. To control time, log it, look for “time leakage” (interruptions, indecision, poor organisation, social media, unmanaged email) and make room for productive work. Working at your biological peak time helps. Covey’s urgent/important quadrants and the 80/20 Pareto principle help with prioritising, and “time stealers” such as waiting and travel can be turned into “time gifters.”

The lectures trace procrastination to Parkinson’s law, fear, laziness, perfectionism and addiction, along with the plain feeling of being overwhelmed. The remedies are:

- Eat the elephant bite by bite.
- Do the “frog” (the hardest task) first.
- Use the Kaizen one-minute principle.
- Keep a clean workspace.
- Set goals you actually want and declare them publicly.
- Reward yourself.
- Use the Zeigarnik effect: once you start, you feel compelled to finish.

The week ends with a first look at assertiveness, or not saying “yes” when you mean “no.”

Week 3: Assertiveness, Anger and Positive Thinking

The lectures compare passive, assertive and aggressive people on body language, listening, feelings and outcomes. Assertiveness means stating your rights without violating anyone else’s, and it ideally ends in a win-win. Practical steps include checking your relationships, dropping self-debilitating thoughts, avoiding discouraging people, following Stanley and Nancy’s commandments and learning four ways to say no.

On anger, Aristotle’s point is that the hard part is being angry at the right person, to the right degree, at the right time. Techniques include postponing your reaction, counting, deep breathing, physical exertion, Lincoln’s unsent letter, silence and channelling anger into creative work. The lectures then turn to positive thinking through the story of two hospital patients and a “window” that faced a blank wall. Changing negative thinking takes desire and determination, introspection and discarding limitations. It also means building a positive self-image through repetition and visualisation, and bringing body, mind and world into sync.

Week 4: People Skills and Being Attractive

Forty traits make others dislike you, and their opposites make people like you. Examples include egotism, aggression, pessimism, fault-finding, gossiping, false promises, moodiness and favouritism, set against selflessness, optimism, giving credit, politeness, trustworthiness, honesty and humility. For dealing with gossip, Socrates' three-filter test (truth, goodness, utility) is suggested.

Attractiveness has physical, mental, emotional and spiritual layers. Physically, it means neat footwear, a fit body, good diet and sleep, and a signature personality. Mentally, it means intellectual charm and a touch of mystery. Emotionally, it means remembering names, valuing yourself and being yourself. Spiritually, it comes down to mindfulness, deeper connection, helping without expectation and "paying it forward."

Week 5: English Skills and Common Errors

English is the global lingua franca, and avoiding common errors is presented as part of soft skills. Using Sonia Sharma's Common Errors in English, the lectures work through repeated self-scored exercises and explanations. Topics include confused word pairs (resume/résumé, principal/principle, alone/lonely), Indianisms (passed out, joint family, cousin brother) and subject-verb agreement (collective, uncountable and plural-form nouns). The lectures also cover prepositions (since/for, at/in, by/on), articles, collocations and word choices like weather/climate. Twain and Branson are quoted on learning by exposure rather than by memorising rules.

Week 6: Humour and Money

Humour reduces stress, improves health and helps communication. It works at three levels: a smile, hearty laughter, and sarcasm, which is best avoided. Adults laugh far less than children (about 4 times a day versus 300). At work, humour resolves conflict, boosts creativity and productivity, and lets you speak the truth lightly. Suggestions include a joke board, a humour club and a humour hall.

The focus then turns to money and personality. How you treat rich and poor people, how you borrow and lend, and how you handle trust all reveal emotional maturity and integrity. Wealth-building advice includes tracking income and expenses, separating need from greed, avoiding impulse buys, and becoming indispensable through devoted, excellent work, with Michelangelo, Martin Luther King and Lincoln as examples. Real wealth also includes things money can't buy.

Week 7: Health and Love

Health affects personality, first impressions and employability, since healthy people are seen as disciplined and reliable. Managing it rests on:

- Regular exercise (morning is best, with tips for sticking to it).
- Diet: regular meals, a heavy breakfast and a light early dinner, fruits and vegetables, about 2 litres of water, and no white foods or soft drinks.
- Sleep: 7-8 hours, a good sleep environment, and avoiding gadgets, late heavy meals and alarms.

Love is defined broadly, drawing on Shakespeare, D. H. Lawrence, Gibran, Lovelace and Corinthians: real love is constant, free, patient and unconditional. Managing love means reading body language and timing, letting relationships mature through stages, handling rejection gracefully and keeping romance separate from work.

Week 8: Ethics, Etiquette, Mind, Memory and Environment

Values, ethics and etiquette make up character ("when character is lost, everything is lost"). Professional ethics combine written codes with society's moral expectations. Poor etiquette and business etiquette are covered in detail: introductions, not interrupting, confidentiality, not criticising your company, and praising in public but reprimanding in private.

The mind is powerful but scattered, and it can be trained through auto-suggestion, positive self-talk and effective thinking (gather facts, sleep on it, trust your gut). Memory techniques include association through absurd images, remembering names and faces, number-image links, memory aids and mental exercise. The final lecture urges care for the environment, with practical steps (save water and electricity, avoid plastic, use public transport, plant trees) and a shift from anthropocentric to eco-centric thinking. It closes with Gandhi's line that the earth provides enough for everyone's need but not everyone's greed.

Experiment No: II-2

SWOC Presentation

Strategic Career Roadmap: SWOC Analysis (slides)

Strategic Career Roadmap: SWOC Analysis

Path to Security Data Scientist / Cybersecurity Analyst

Candidate: B Idhayachandran | 1st Year B.E. CSE | Batch 2026 – 2030

Goal: build strong fundamentals in security and data early, then combine them into threat-detection and analytics work

Executive Summary & Career Vision

1 Vision

Start as a first-year CSE student and grow into a professional who uses data science to detect and prevent cyber threats.

2 Objective

Learn Linux, networking and Python properly, then security and data analysis. Put judgment, evidence and problem-solving ahead of memorised tools.

3 Framework

Use a SWOC matrix (Strengths, Weaknesses, Opportunities, Challenges) to draw up a 4-year milestone roadmap.

SWOC Matrix at a Glance

Strengths (Internal)	Weaknesses (Internal)
Curiosity about how systems fail; steady Python and C practice; strong interest in both security and data	Beginner in networking, operating systems and statistics; no certifications or internships yet
Opportunities (External)	**Challenges (External)**
Growing demand for security analysts and data professionals; AI-driven threat detection is a fast-growing area	Fast-changing threats and tools; entry roles ask for experience; practice must stay legal and ethical

Strengths

1	Curious problem solver	I enjoy working out why something does not behave as expected, which is the core habit of both security analysis and data science.
2	Programming foundation	Practice in Python and C through coursework and projects.
3	Two-field interest	I can connect security work with data analysis, such as spotting unusual activity in logs.
4	Willingness to learn	I am ready to study fundamentals first and tools later.

Weaknesses

Experience gaps

I am a beginner in networks, operating systems, databases and statistics. I have no internship or certification so far.

Working style

I may rush to try tools before I understand the underlying idea. Security and data work both reward planning, notes and careful verification.

Opportunities

Rising demand

Organisations need people who can monitor systems, analyse incidents and work with large data sets.

AI in security

Machine learning is increasingly used for anomaly detection, fraud detection and malware analysis.

Free learning platforms

Beginner-friendly practice sites, open datasets, Kaggle and student programs provide hands-on learning.

Community

Hackathons, capture-the-flag (CTF) events, college clubs and meetups help build skills and contacts.

Challenges

Fast-changing field

New threats, tools and frameworks appear constantly, so learning has to be continuous.

Experience paradox

Entry roles often ask for experience that students have had little chance to gain.

Ethics and law

Security skills must be practised only on systems I own or have permission to test, such as practice labs.

Data quality

Poor or biased data leads to wrong conclusions, so I need to learn to check it.

Action Plan: Closing the Gaps

1 Working style

Before each project, write a short note: goal, data or tools, risks. Then build.

2 Knowledge gap

Use Linux regularly. Study networking basics (TCP/IP, DNS, ports), SQL and basic statistics alongside coursework.

3 Credential gap

Complete an entry-level security or data certificate before the end of Year 2.

Action Plan: Using Opportunities

Build projects

A log-analysis notebook, a password-strength checker and a small anomaly-detection project, each with a README.

Practice legally

Use beginner CTFs and safe practice labs only.

Use AI as a tool

Let it draft boilerplate while I focus on understanding, review and testing.

Join communities

Take part in hackathons, college clubs and student security events.

4-Year Milestones Matrix

Year	Primary Focus	Target Deliverable	Target Credential
Year 1 (2026–27)	Linux, networking, Python, SQL basics	GitHub portfolio with a log-analysis notebook	Entry-level cybersecurity or data certificate
Year 2 (2027–28)	Statistics, machine learning, web and network security	Anomaly-detection project on a public dataset	CompTIA Security+ or a similar certificate
Year 3 (2028–29)	Threat detection, SIEM tools, cloud security	Mini security-monitoring dashboard	Cloud security or data analyst certificate
Year 4 (2029–30)	Security analytics, leadership	Capstone: ML-based threat detection; team lead	Placement / off-campus hiring

Conclusion & Key Takeaways

1 Think first, then build

Planning before coding turns curiosity into reliable work.

2 Fundamentals plus AI

AI-generated code and models are trustworthy only when someone understands the systems and data beneath them.

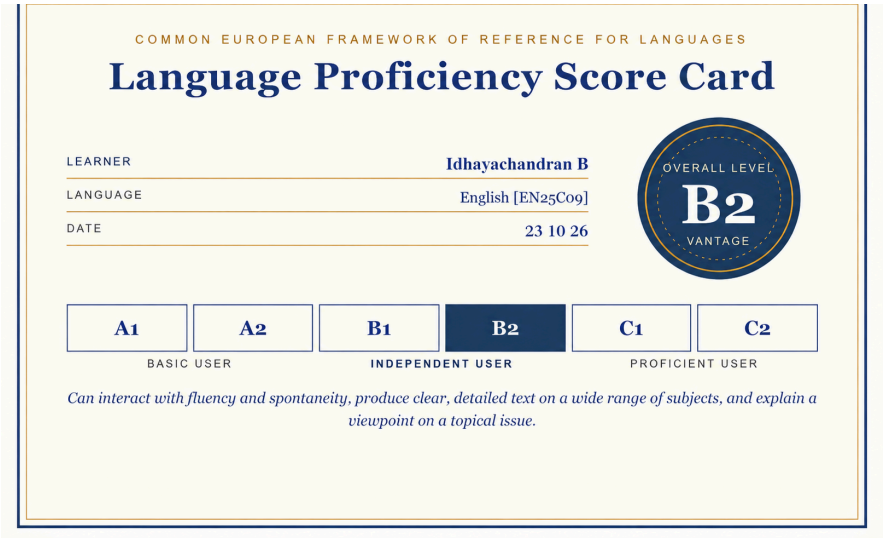
3 Verdict

With disciplined study and steady hands-on practice, I can build a profile that combines cybersecurity and data science.

Experiment No: II-3 & 4

Write and Improve / Speak and Improve

Write and Improve: CEFR score card



CEFR score card

Learner: B Idhayachandran Language: English [EN25C09] Date: 23 10 26

A1	A2	B1	B2	C1	C2

My CEFR level is B2 (Independent User).

Result shown in the score card above.

Experiment No: II-5

Professional English Practice Worksheets – Solved

Course Code: EN25C09 | Subject: Professional English – I (Lab Integrated) | Worksheet 1: Model Exam I (Parts A, B, C), followed by Worksheet 2: Model – I Practice Worksheet.

Worksheet 1 (EN25C09 Model Exam I) – Part A (15 × 1 = 15 Marks)

1. monitors
2. recorded
3. stronger
4. must
5. can
6. The laboratory developed a new testing device that measures material strength with high precision.
7. rises
8. would not have failed
9. The mechanics installed dual damper springs to isolate chassis resonance.
10. optimizing
11. 50 mL of hydrochloric acid was poured into the titration flask.
12. smoothly
13. The project supervisor asked the engineer whether he had calibrated the pressure sensor before testing.
14. The incorrect noun form is "equipments", because "equipment" is uncountable. Correct sentence: The laboratory purchased several new pieces of equipment for conducting material testing.
15. worse

Worksheet 1 – Part B (10 × 2 = 20 Marks)

1. The engineer calibrated the pressure sensor and verified its accuracy.
2. (Portability) A laptop is more portable than a desktop computer because it is lighter and can be carried anywhere. (Screen Size) A desktop computer usually has a larger screen than a laptop, so it is more comfortable for long hours of detailed work.
3. The engineer designed a new cooling system, which reduces the temperature of industrial machines.
4. (a) If the battery voltage drops below the required level, the device will shut down automatically. (b) If the technician had checked the connections earlier, the fault would have been detected before the breakdown.
5. The solution was heated to 80°C before testing.
6. The project manager asked the technician whether he had inspected the pressure valves before starting the compressor.
7. (a) The engineer received a lot of feedback on the new design. ("Feedback" is uncountable, so it cannot take "many" or the plural form "feedbacks".) (b) The students received several pieces of useful advice from their teacher. ("Advice" is uncountable, so "advices" is wrong.)
8. (a) The technician wears safety gloves to protect his hands. (b) The cooling fan operates continuously to prevent the motor from overheating.
9. (a) efficiently (b) carefully
10. (a) can (b) must not

Worksheet 1 – Part C (5 × 13 = 65 Marks)

Question 26 (a) Safety instructions for safe and responsible use of the internet

1. Use a strong, unique password of at least twelve characters, mixing letters, numbers and symbols, for every account.
2. Enable two-factor authentication on your email, banking and social media accounts.
3. Do not click on links or open attachments from unknown or suspicious senders.
4. Never share personal details such as OTPs, passwords, Aadhaar or bank details with anyone online.

5. Check that a website address begins with "https" and shows a padlock before entering any sensitive information.
6. Keep your operating system, browser and antivirus software updated at all times.
7. Avoid banking or shopping on public Wi-Fi; use a trusted VPN if you must connect.
8. Report cyberbullying, online fraud and phishing attempts immediately to a trusted adult, the platform, or the National Cyber Crime Helpline (1930).

Question 27 (a) Descriptive essay: A traditional art form in my region and how modern initiatives can preserve it

Bharatanatyam: The Dance That Carries Tamil Nadu's Soul

Among the many traditional art forms of Tamil Nadu, Bharatanatyam is the one that has carried our culture across the centuries. It began inside the temples, where devadasis performed it as an offering to the deity. The epic Silappathikaram describes classical dance in detail, and the carvings on the Brihadeeswarar Temple at Thanjavur and the Chidambaram Nataraja Temple show its postures in stone. In the nineteenth century, the Thanjavur Quartet organised the repertoire into the form we see today.

Traditionally, a dancer trains for many years under a guru. A performance blends nritta (pure rhythm), nritya (expressive storytelling) and natya (drama), told through hand gestures called mudras, facial expressions and Carnatic music. The dancer wears a bright silk costume, temple jewellery and ankle bells called salangai. The first full solo performance, the arangetram, marks the end of a student's formal training.

Today, many young people feel the art form is old-fashioned and too expensive to learn. Modern initiatives can change this. Schools and colleges can include classical dance in cultural clubs and the curriculum, as institutions like Kalakshetra in Chennai already do. Online classes, recorded performances and social media reels can reach learners in small towns. Festivals such as the Margazhi season in Chennai give young artists a stage, and scholarships can help families who cannot afford training.

Bharatanatyam is more than entertainment; it is a living record of our history, music and devotion. If we support it through education, technology and public platforms, the next generation will carry the dancing bells forward.

Question 28 (b) Formal letter of request to the Principal

B Idhayachandran
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Principal
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission to attend a technical workshop on Artificial Intelligence and Emerging Technologies

Respected Sir/Madam,

I am a first-year B.E. Computer Science and Engineering student of your college. I have come to know that [Name of Institution] is organising a one-day technical workshop on Artificial Intelligence and Emerging Technologies on [date]. I request your kind permission to attend this workshop.

My objective in attending is to learn how artificial intelligence is applied in real engineering problems and to gain hands-on exposure to tools that are not covered in detail in our first-year syllabus. I also hope to meet experts and students from other institutions and learn from their experience.

I expect to learn the basics of machine learning, practical uses of AI in software and cloud systems, and the latest trends in emerging technologies. After the workshop, I will share the notes and key points with my classmates and use what I learn in my Python projects.

I request you to kindly grant me on-duty leave for the day of the workshop and permit me to travel to the venue. I will make up for any classes missed and will submit the participation certificate to the department after the workshop.

I would be grateful if you could grant me this permission. Thank you for your kind consideration.

Yours obediently,

B Idhayachandran
(Register No.: 26CS1264)

Question 29 (a) Analytical essay: AI-based assessment versus traditional teacher-based assessment

AI-Based Assessment versus Traditional Teacher-Based Assessment in Engineering Education

Assessment decides how students learn, so the choice between AI-based and teacher-based methods matters in engineering education. Each has clear strengths and weaknesses.

In terms of accuracy, AI tools are consistent. An automated system can evaluate hundreds of code submissions or objective answers using the same rules every time, without getting tired. Teachers, on the other hand, can understand a student's reasoning in a long derivation or a design report, which software often misses. Therefore, AI is accurate for objective answers, whereas teachers are better for open-ended ones.

Feedback time is where AI wins most clearly. Students receive results and error explanations within seconds, while manual evaluation of a large class may take weeks. Quick feedback lets students correct their mistakes while the topic is still fresh. Conversely, a teacher's comments arrive more slowly but are usually more personal and motivating.

Regarding fairness, AI does not show favouritism or mood-based marking. However, it can carry hidden bias from the data it was trained on, and it may penalise students who give unusual but valid solutions. A teacher can notice such cases and adjust the marks.

Finally, human judgement remains important. Engineering involves creativity, ethics and teamwork, which cannot be reduced to a score.

In conclusion, neither method is perfect alone. The best approach is a blend, where AI handles routine and objective checking quickly, and teachers review complex work and guide students. This combination keeps assessment fast, fair and meaningful.

Question 30 (b) Personal email: Orientation programme

To: [friend's email address]
Subject: My first-year orientation programme at college

Dear [Friend's name],

How are you? I hope you are doing well and enjoying your college life too. I have been wanting to write to you about my orientation programme, so here it is.

The programme was held for the first-year engineering students at our college. It began with a welcome address by the Principal, followed by talks from senior faculty and some alumni. We also had fun icebreaker games and team activities, which helped me make new friends within the first day itself.

The best part was the tour of the academic facilities. We visited the library, the computer labs, the workshops and the seminar halls, and the staff explained how to use each of them. We were also told about the college rules on attendance, discipline, dress code and examinations, so now I know exactly what is expected from us.

Overall, the programme made me feel much more comfortable in the new environment. I understood how college life works and how to plan my studies, and I felt ready to start my academic journey with confidence.

Write back soon and tell me about your college too. Take care!

Your friend,
Idhaya

Worksheet 2 (Model – I Practice Worksheet) – Exercises 1 to 15

Exercise 1: Simple Present Tense (Subject-Verb Concord)

1. recognize
2. requires
3. transmit
4. agrees

5. form

Exercise 2: Simple Past Tense

1. conducted
2. deployed
3. opened
4. published
5. detected

Exercise 3: Comparative Adjectives

1. higher
2. safer
3. faster
4. more durable
5. lower

Exercise 4: Modals of Obligation and Necessity

1. must
2. must
3. have to
4. must
5. has to

Exercise 5: Modals of Capability and Possibility

1. can
2. may
3. can
4. can
5. can

Exercise 6: Relative Clauses

1. The civil engineer who designed the suspension cable bridge received the national infrastructure award.
2. The spectrometer, which analyzes infrared emission spectra, is mounted inside the cleanroom vacuum chamber.
3. The technical specialist whose patent on pneumatic grippers was approved was reassigned to the robotics lab.
4. This microcontroller, which features ultra-low power sleep states, is ideal for battery-operated remote sensors.
5. The field researchers who surveyed the wind turbine farm submitted their acoustic emission assessment yesterday.

Exercise 7: If-Clauses (Type 1)

1. exceeds
2. fails
3. engages
4. rise
5. registers

Exercise 8: If-Clauses (Type 3)

1. had started
2. had lubricated
3. would have been prevented
4. would have tripped
5. had remained

Exercise 9: Infinitives of Purpose

1. The mechanics installed dual damper springs to isolate chassis resonance.
2. Technicians purge the fuel line with dry nitrogen gas to remove residual flammable vapors.
3. We introduced an optical encoder to determine spindle velocity accurately.
4. The developers wrote an automated load testing script to simulate ten thousand concurrent connections.
5. Gold plating is applied to terminal connectors to prevent atmospheric corrosion.

Exercise 10: Gerunds after Prepositions

1. redesigning
2. replacing
3. reducing
4. cleaning
5. filtering

Exercise 11: Impersonal Passive Voice

1. 50 mL of hydrochloric acid was poured into the titration flask.
2. The robotic arm was tested under a 50 kg dynamic payload.
3. The optical pyrometer was calibrated prior to ignition.
4. Severe cavitation damage was observed along the impeller blades.
5. Volatile solvents are stored in flame-resistant containment cabinets.

Exercise 12: Adverbs of Manner

1. precisely
2. periodically
3. rapidly
4. firmly
5. smoothly

Exercise 13: Reported Speech

1. The quality manager asked the technician whether the insulation resistance test had been documented.
2. The lead architect enquired whether the server cluster could handle double the transaction throughput.
3. The field auditor asked the safety engineer whether he had inspected the gas leakage relief lines.
4. The client asked the team whether the firmware update would fix the intermittent battery drain.
5. The instructor asked the lab group why the multimeter had shown fluctuating voltage readings.

Exercise 14: Countable and Uncountable Nouns

1. Incorrect: "machineries" and "equipments". Correct: The procurement officer purchased several new pieces of machinery and equipment for the workshop.
2. Incorrect: "datas". Correct: The laboratory generated a huge amount of test data during the wind tunnel trials.
3. Incorrect: "softwares". Correct: We need three more software licenses to complete the finite element analysis on the model.
4. Incorrect: "informations". Correct: The field report provided a lot of valuable information regarding the soil composition.
5. Incorrect: "apparatuses" and "glasswares". Correct: All chemical apparatus and glassware must be washed thoroughly after the experiment.

Exercise 15: Comparative Adjectives (Irregular Forms)

1. worse
2. better
3. farther
4. less

5. further

Worksheet 2 – Part B (Exercises 1 to 10)

Exercise 1: Simple Past Tense Conversion

1. The technician inspected the solar arrays and recorded the peak power conversion value.
2. The chemical analyzer detected trace mercury levels and sent an automated alert to the environmental dashboard.
3. Maintenance crews lubricated the heavy conveyor bearings and replaced the worn rubber belts before the morning shift.
4. The supervisory software polled field sensors every five milliseconds and archived temperature trends into a secure database.
5. The testing team applied a 500-volt dielectric pulse across the stator winding and monitored for insulation breakdown.

Exercise 2: Comparative Sentences

1. Cloud storage has a lower initial deployment cost than on-premise physical servers because no hardware has to be bought. It is also more elastic, since its capacity can be increased within minutes.
2. Fiber-optic cables offer a much higher data transfer bandwidth than copper twisted-pair cables. They are also less susceptible to electromagnetic interference than copper cables.
3. Electric vehicles produce lower tailpipe emissions than internal combustion engine vehicles, as they release none at all. However, EVs generally have a higher initial purchase price than ICE vehicles.
4. Hydraulic actuators produce a greater raw force output than electromechanical actuators. They also carry a higher risk of fluid leakage, because electromechanical actuators use no fluid.
5. Relational (SQL) databases have a more rigid schema than document (NoSQL) databases. Document databases usually provide higher horizontal query throughput than relational databases.

Exercise 3: Relative Clauses

1. The algorithm that analyzes microscopic cracks in turbine blades was patented by the aeronautics team.
2. The lead chemical engineer, who noticed an abnormal pressure spike in chamber two, authorized the emergency boiler shutdown.
3. The laboratory houses a 400 kV impulse voltage generator, where high-voltage insulator puncture tests are conducted.
4. We engaged an external cybersecurity consultant whose firmware vulnerability audit exposed three backdoor ports.
5. The supervisory system, which utilizes an artificial neural network, predicts transformer insulation life based on dissolved gas concentrations.

Exercise 4: If-Clauses

1. (a) If the operating temperature exceeds 85°C, the cooling system will switch on automatically. (b) If the operator had pressed the emergency shutdown switch immediately, the accident would have been avoided.
2. (a) If the supply voltage drops below the 200V threshold, the backup power supply will take over. (b) If the network administrator had enabled multi-factor authentication, the unauthorized access would have been prevented.
3. (a) If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel, the ventilation fans will start and an alarm will sound. (b) If the mechanical team had checked the bearing lubrication schedule, the bearing would not have overheated.
4. (a) If the autonomous guided vehicle (AGV) detects an obstacle within 1 meter, it will stop immediately. (b) If the thermal sensor had registered the runaway reaction in time, the explosion would have been prevented.
5. (a) If the software receives an unauthenticated API token, it will reject the request and return an error message. (b) If the civil inspectors had reviewed the core sample density tests, they would have detected the weak foundation before construction.

Exercise 5: Impersonal Passive Voice

1. The crude extract was distilled at a constant temperature of 60°C.
2. The cross-section of the weld seam was polished with fine diamond paste.
3. Fifteen cast-iron pump casings were rejected due to internal porosities.
4. The resistance across the bridge circuit was measured using a calibrated digital ohmmeter.
5. The emergency aerodynamic braking flaps were initiated at an altitude of 3,000 meters.

Exercise 6: Reported Speech

1. The safety supervisor asked the intern whether he had verified the grounding connections on the generator.
2. The plant director asked the site engineer whether the emergency effluent treatment plant had received clearance.
3. The IT security lead enquired of the developer why he had disabled the firewall rule on the database server.
4. The client asked the project manager whether the bridge structure could withstand Category 4 hurricane winds.
5. The research guide asked the postgraduate scholar how many sample trials he had conducted under cryogenic conditions.

Exercise 7: Countable vs. Uncountable Nouns

1. (a) We need to install several new pieces of software and machinery in the CAD laboratory. (b) The team collected plenty of useful information from the field test.
2. (a) The technician ordered ten new pieces of apparatus and modern electronic equipment for the antenna lab. (b) High-speed train operations require advanced transport infrastructure and dedicated electrical power.
3. (a) The environmental sensor gathers extensive atmospheric data and pollution indices every minute. (b) The construction crew unloaded tons of sand, cement, and steel debris at the worksite.
4. (a) All electrical hardware and wiring must comply with local safety legislation. (b) The data scientist discovered several interesting insights after running the machine learning clustering.
5. (a) Laboratory staff must clear away all glassware and acidic waste before leaving the premises. (b) The project failed due to a lack of proper funding and technical research.

Exercise 8: Infinitives of Purpose and Gerund Constructions

1. (a) A pressure relief valve is installed to vent excess boiler steam. (b) The automated script runs every midnight to archive legacy server logs.
2. (a) Automotive engineers installed an intercooler to reduce the compressed air intake temperature. (b) The technicians analyzed the structural stress points by placing piezoelectric strain gauges along the girder.
3. (a) High-performance heatsinks feature copper heatpipes to conduct heat away from the silicon die. (b) The factory cut water consumption by 30% by implementing a closed-loop recycling filtration system.
4. (a) Safety guards are mounted over rotating flywheel belts to prevent accidental worker contact. (b) The algorithm optimizes memory bandwidth by pre-fetching sequential array indices into L2 cache.
5. (a) The electronics lab installed an uninterruptible power supply (UPS) to protect servers against sudden voltage sags. (b) The metallurgy group enhanced surface hardness by diffusing nitrogen atoms into the steel casing.

Exercise 9: Adverbs of Manner

1. (a) automatically (b) flawlessly
2. (a) accurately (b) instantly
3. (a) continuously (b) cautiously
4. (a) consistently (b) catastrophically
5. (a) periodically (b) smoothly

Exercise 10: Modals

1. (a) can (b) must not

2. (a) must (b) may
3. (a) must not (b) could
4. (a) should (b) must
5. (a) can (b) must not

Worksheet 2 – Topic 1: Writing Imperative Safety Instructions

Question 1: Chemistry and Materials Testing Laboratory

1. Wear a lab coat, safety goggles and nitrile gloves at all times inside the laboratory.
2. Handle concentrated acids and alkalis only inside the fume hood.
3. Add acid to water slowly; never add water to acid.
4. Do not smell, taste or touch any chemical directly.
5. Label every container clearly, and never return unused chemicals to the stock bottle.
6. Clean all glassware and testing equipment after each experiment to avoid contamination.
7. Keep food, drinks and open flames away from the work area.
8. Report every spill, burn or breakage to the instructor at once, and use the eyewash station or safety shower in an emergency.

Question 2: High-Voltage Electrical Machines and Drives Laboratory

1. Make sure the main supply is switched off before making or changing any connection.
2. Wear insulated rubber gloves and safety shoes, and stand on an insulated rubber mat near live panels.
3. Never touch exposed terminals, bare wires or rotating parts while the machine is running.
4. Get the circuit connections checked by the instructor before switching on the supply.
5. Know the location of the circuit breakers, fuses and emergency stop button, and keep them in working condition.
6. Remove rings, watches and metal jewellery, and avoid loose clothing, to prevent arc flash burns and accidental short circuits.
7. Keep water and wet hands away from electrical equipment, and use only insulated tools.
8. Increase the motor load gradually, and switch off the supply and discharge the capacitors before disconnecting the circuit.

Question 3: Advanced Manufacturing and CNC Machining Workshop

1. Wear safety glasses, safety shoes and close-fitting clothes, and tie back long hair.
2. Keep the machine door and guard closed while the spindle is running.
3. Do not touch the spindle, cutter or workpiece until the machine has stopped completely.
4. Clamp the workpiece and tool holder firmly before starting the program.
5. Use a brush or chip hook, and heat-resistant gloves, to remove hot metal chips, and only after stopping the machine.
6. Wipe up coolant spills immediately, and keep the floor dry to prevent slipping.
7. Run the program in simulation mode first, and keep your hand near the emergency stop button during the first cut.
8. Never leave the machine unattended or operate it without the instructor's permission.

Worksheet 2 – Topic 2: Free Writing – Festivals and Cultural Celebrations

Question 1: A traditional harvest festival and eco-friendly practices

Pongal: The Harvest Festival of Tamil Nadu and How Eco-Friendly Practices Can Protect Its Spirit

Pongal is the most important harvest festival of Tamil Nadu. It is celebrated in mid-January to thank the Sun God, the rain and the cattle for a good harvest. The festival lasts four days: Bhogi, Thai Pongal, Mattu Pongal and Kaanum Pongal.

On Bhogi, families clean their homes and throw away old things. On Thai Pongal, they draw colourful kolams at the entrance and cook fresh rice and milk in a new clay pot in the open yard. When the milk

boils over, everyone shouts "Pongalo Pongal!" The pot is decorated with turmeric plants and sugarcane. The festive meal includes sakkarai pongal, venn pongal, sambar, vadai and payasam, served on banana leaves. On Mattu Pongal, farmers decorate and worship their cattle, and Kaanum Pongal is a day for family visits and community games.

Modern eco-friendly practices can preserve the core spirit of the festival. Instead of burning plastic and old tyres on Bhogi, communities can burn only dry leaves or turn them into compost, which reduces smoke and air pollution. Clay pots, banana leaves and areca-leaf plates can replace plastic and thermocol, and kolams can be drawn with natural rice flour. Villages can organise tree-planting drives, rainwater harvesting and cloth-bag campaigns during the festival, and use solar lamps instead of excess electric decorations.

In this way, Pongal can remain joyful and traditional while staying kind to nature, and the next generation will learn that celebration and conservation can go together.

Question 2: A temple car festival and sustainable measures

The Temple Car Festival: Faith, Food and a Cleaner Celebration

In my hometown, as in many towns of Tamil Nadu, the temple car festival, or Ther Thiruvizha, is the biggest event of the year. Thousands of devotees gather in the streets to watch the deity pulled in a huge, richly decorated wooden chariot, called the ther.

The festival begins with flag hoisting and several days of processions. On the car day, devotees of all ages hold thick ropes and pull the heavy chariot together through the four main streets, chanting prayers while drums and nadaswaram fill the air. This shared effort is a strong symbol of unity, because people from every community take part. Women draw kolams on the road, and families offer coconuts, bananas and flowers to the deity. Volunteers give free food such as pongal, puliyodharai, curd rice and sweet payasam as prasadam, served on leaves.

However, such a large crowd also creates problems. Single-use plastic cups, plates and bags litter the streets, and loudspeakers cause noise pollution. Eco-friendly measures can solve these problems. Organisers can ban single-use plastic and provide steel tumblers, banana leaves and paper bags. Dustbins and waste-sorting points can be placed along the route, and volunteers can collect and compost the food waste. Noise can be controlled by limiting loudspeaker volume and using traditional instruments. Barricades, marked entry and exit lanes, volunteers, and first-aid points can manage the crowd safely.

With these steps, the festival can keep its sacred joy and remain clean and peaceful.

Question 3: Regional folk art and handicraft fairs with renewable energy and zero-waste practices

Folk Art and Handicraft Fairs: Heritage with a Green Future

A regional folk art and handicraft fair is a colourful meeting of tradition and community. Walking through the fair grounds, visitors see Karagattam and Oyilattam performances, hear the beat of the parai drum, and watch artisans make Thanjavur dolls, palm-leaf baskets, clay pots and silk sarees. The air smells of fresh filter coffee, murukku, adhirasam and hot sundal.

These fairs are important because they give artisans a market and help young people see the value of their heritage. Many stalls offer live demonstrations, so children can try pottery or weaving themselves. Local delicacies prepared by village cooks add to the festive atmosphere and support small businesses.

However, large fairs use a lot of electricity and produce a lot of waste. Modern environmental strategies can solve this. Solar micro-grids, made of rooftop panels and batteries, can power the lights, sound systems and stalls without diesel generators. Zero-waste composting can turn leftover food and leaves into manure for local farms. Organisers can use reusable steel plates, clay cups and cloth banners, provide separate bins for wet and dry waste, and recycle paper and cardboard.

If we combine the traditional skill of our artisans with clean energy and responsible waste management, these fairs can protect our culture and our planet together.

Worksheet 2 – Topic 3: Free Writing – Food and Technical Travel Plans

Question 1: Technical study tour to a major hydroelectric dam

A Technical Study Tour to the Mettur Hydroelectric Dam

Our department has planned a one-day technical study tour to the Mettur Dam on the Cauvery River in Salem district. The group of forty students and two faculty members will leave the college by a chartered bus at 4:30 a.m. and travel along the national highway, with a short breakfast halt on the way. The journey takes about six hours, and we will return by night. Students will carry ID cards, water bottles and a notebook.

After registering at the gate, an engineer will take us through the powerhouse. The long concrete dam holds back the huge Stanley Reservoir. Inside the powerhouse, we will see the penstocks, the turbines and the generators, and learn how falling water turns turbines to produce electricity. We will also observe the control room, where operators monitor water level, flow and power output on screens, and the spillway gates used to release extra water safely.

For lunch, we will try local dishes such as hot rice with sambar, fish fry from the river and fresh mango juice, which make the trip a cultural experience as well. In the evening, we will walk through the nearby park and enjoy the green landscape and the calm reservoir.

The visit will help us understand renewable energy, water management and the role of civil, mechanical and electrical engineering in one project. We will follow all safety rules, wear helmets where needed, and stay with our group at all times.

Question 2: Technical study tour to an integrated solar and wind hybrid energy farm

A Technical Study Tour to a Solar and Wind Hybrid Energy Farm

We plan a two-day study tour to the Muppandal wind farm region near Kanyakumari, where wind turbines and solar panels work together. Thirty students and two faculty members will take the overnight train from Chennai to Nagercoil, then travel about one hour by a hired van to the farm. We will stay one night in a budget lodge nearby, and return by train on the second night.

On arrival, a site engineer will give a short briefing on how the hybrid plant works. The wind turbines turn on tall towers, and their blades rotate a generator inside the nacelle. We will watch how the strong coastal wind drives them, and how the blades adjust their angle with the wind speed. In the solar section, we will observe rows of photovoltaic panels, inverters and transformers that feed power into the grid. The control room will show how the two sources balance each other: solar power peaks in the afternoon, whereas wind is strong in the evening and night.

For food, we will taste Nanjil Nadu specialities such as fish curry with rice, banana chips, avial and fresh coconut sweets. Eating in small local hotels along the route will also help us understand the region's culture.

The tour will show us the practical side of renewable energy, and we will follow safety rules by wearing helmets, keeping away from the turbine bases and obeying the engineers' instructions.

Question 3: Technical field visit to an automated seawater reverse osmosis desalination plant

A Technical Field Visit to a Seawater Reverse Osmosis Desalination Plant

Our class is planning a half-day field visit to a seawater reverse osmosis desalination plant on the East Coast Road near Chennai, such as the Nemmeli plant. About fifty students and three faculty members will travel in two college buses, leaving at 8:00 a.m. and reaching the plant in about ninety minutes. We will carry ID cards and wear closed shoes for safety.

At the plant, the engineers will first explain the process. We will see the intake pumping station, which draws seawater through long pipes under the sea bed. The water then passes through pre-treatment filters that remove sand and suspended particles. Next, high-pressure pumps push the water through the membrane filtration racks, where the salt is separated and fresh water is produced. We will also see the brine discharge outlet, where the concentrated salt water is carefully diluted and released back into the sea to protect marine life. The control room shows the whole process through automated screens.

After the visit, we will have lunch at a local seaside restaurant. The coastal menu includes fish fry, prawn masala, crab curry with steamed rice and filter coffee, which gives us a taste of the Chennai coastal culture.

The visit will help us see how engineering solves water scarcity, how energy is used in the process, and why environmental care in brine disposal is important.

Worksheet 2 – Topic 4: Formal Letters of Request

Question 1: Letter to the Head of the Computer Science Department (lab access for a hackathon)

B Idhayachandran
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Head of the Department
Department of Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission and lab access after college hours to prepare for a National AI Hackathon

Respected Sir/Madam,

I am writing on behalf of my project team of four first-year students. We have registered for a National AI Hackathon to be held on [date], and we respectfully request permission to use the computer laboratory after college hours.

The objective of our project is to build a simple AI-based solution that solves a real campus problem, using Python and machine learning. We need about three weeks to design, train and test the model, and our home systems are not powerful enough for this.

We require access to the lab between 4:30 p.m. and 7:30 p.m. on weekdays, with Python, Jupyter Notebook, TensorFlow or scikit-learn, Git and a stable internet connection. We promise to follow all lab rules, maintain the equipment carefully, and leave the lab clean and switched off after use. We will also work under the supervision of a faculty member if required.

We would be very grateful if you could grant us this permission. Thank you for your kind consideration.

Yours obediently,

B Idhayachandran
(On behalf of the project team)

Question 2: Letter to the Plant Manager (industrial plant visit)

B Idhayachandran
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Plant Manager
[Name of Factory], Industrial Robotics and Automation Division
[Address]

Subject: Request for permission to conduct an industrial visit for 50 first-year engineering students

Respected Sir/Madam,

I am a first-year engineering student of St. Joseph's College of Engineering, Chennai. On behalf of my class, I request your permission to visit your robotics and automation factory with fifty students and two faculty members on a convenient date in [month].

The intended learning outcomes of the visit are to understand how robots and automated systems work on a real production line, to see how sensors, controllers and robotic arms are combined, and to learn about industrial safety practices and quality control. The visit will also help us connect our classroom learning with practical engineering and inspire us to think about careers in automation.

We will arrive in a group, carry our ID cards, follow all safety rules, and obey your staff's instructions throughout. We would be glad if a short briefing and a guided tour of about two hours could be arranged.

I would be grateful if you could kindly grant us this opportunity. I look forward to your reply.

Yours faithfully,

B Idhayachandran

Question 3: Letter to the Head of the Mechanical Engineering Department (machine shop access for an electric go-kart)

B Idhayachandran
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Head of the Department
Department of Mechanical Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission and machine shop access after college hours to build an electric go-kart

Respected Sir/Madam,

Our student team has been selected to take part in an inter-collegiate design challenge to be held on [date], where we must build and test an energy-efficient electric go-kart. We therefore request permission to use the machine shop after regular college hours.

We need the workshop to cut and weld the chassis, fit the motor, battery and braking system, and test the kart on the campus ground. We estimate that we will need about four weeks, working from 4:30 p.m. to 7:30 p.m. on weekdays and during part of the weekends.

We will work only under the supervision of the workshop instructor, wear safety glasses, gloves and shoes at all times, and follow all machine safety rules. We will take care of the tools and keep the shop clean after use.

We would be thankful if you could grant us this permission and allow us to bring honour to the college. Thank you for your kind support.

Yours obediently,

B Idhayachandran
(On behalf of the team)

Question 4: Letter to the Principal / Dean of Student Affairs (sponsorship and duty leave)

B Idhayachandran
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Principal / Dean of Student Affairs
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for financial sponsorship and duty leave to present a drone surveillance prototype at the National Defense Innovation Expo

Respected Sir/Madam,

I am happy to inform you that a five-member multidisciplinary team of our college has been selected to present a drone surveillance prototype at the National Defense Innovation Expo, to be held on [dates] at [venue]. We respectfully request financial sponsorship and duty leave for this event.

Our prototype uses a camera-equipped drone with image processing to monitor sensitive areas in real time. Presenting it at a national level will give us valuable feedback from experts and will bring recognition to our college.

We request sponsorship for travel, accommodation, registration and prototype transport, which we estimate at ₹[amount], and duty leave for [number] days so that we do not lose attendance. We will make up for the missed classes and submit a report after the event.

I would be grateful for your kind support and approval. Thank you.

Yours obediently,

B Idhayachandran
(On behalf of the team)

Worksheet 2 – Topic 5: Compare and Contrast Descriptive Essays

Question 1: Electric Vehicles versus Internal Combustion Engine Vehicles

Electric Vehicles versus Conventional Internal Combustion Engine Vehicles

Electric vehicles (EVs) and internal combustion engine (ICE) vehicles are two different answers to the same need, which is transport. Each has clear positives and negatives.

Consider battery life first. An EV battery usually lasts eight to ten years, but replacing it is very expensive. In contrast, an ICE vehicle has no battery problem, though its engine needs regular servicing. As for the manufacturing footprint, EVs create more emissions during production because of battery mining and processing, whereas ICE vehicles have a lighter manufacturing footprint. However, over its lifetime an EV produces far lower emissions, especially when the electricity comes from renewable sources.

Operating costs also differ. EVs have fewer moving parts, so they need less maintenance, and electricity per kilometre is cheaper than petrol or diesel. On the other hand, ICE vehicles cost less to buy, and their repair shops are available everywhere. Fuel prices, however, keep rising and make them costly to run.

Charging network feasibility is the biggest challenge for EVs. Charging stations are still limited, especially on highways and in small towns, and charging takes much longer than refuelling. ICE vehicles, conversely, can be refuelled in minutes at petrol stations spread across the country.

In conclusion, EVs are cleaner and cheaper to run, whereas ICE vehicles are cheaper to buy and more convenient at present. As battery prices fall and charging networks grow, EVs are likely to become the better choice for the future.

Question 2: Remote/Virtual Internships versus On-Site Internships

Remote Internships versus On-Site Internships for First-Year Engineering Students

Internships help early-stage engineering students connect classroom theory with real work. Today, students can choose between remote (virtual) and on-site (physical) internships, and each has advantages and disadvantages.

Hands-on skill acquisition is the biggest difference. On-site internships give direct access to machines, laboratories and real equipment, so students learn practical skills such as wiring, fabrication and testing. Remote internships, in contrast, mostly involve software, simulations and online tasks, which are useful for coding and design but cannot replace physical practice.

In terms of schedule flexibility, remote internships win clearly. Students can work from home, save travel time and cost, and manage studies alongside the internship. On-site internships, on the other hand, demand fixed hours and daily travel, which can be tiring during the semester.

Mentorship is stronger on-site, because supervisors can guide students directly and correct mistakes immediately. Online mentors, however, are often busy and communicate only through scheduled calls or messages, so doubts may stay unresolved for days.

Professional networking also favours on-site internships, as students meet colleagues, build relationships and learn workplace culture naturally. Remote interns may never meet their team face to face, although online platforms still allow them to connect with professionals from different cities.

In conclusion, remote internships are better for flexibility and software skills, whereas on-site internships are better for practical learning, mentorship and networking. A first-year student should ideally begin with a remote internship and move to an on-site one later.

Question 3: Cloud Computing Infrastructure versus On-Premise Data Centers

Cloud Computing versus On-Premise Physical Data Centers for Engineering Enterprises

Engineering enterprises must decide whether to run their systems on the cloud or in their own physical data centers. Both choices have strengths and weaknesses.

Upfront capital expenditure is the clearest difference. Cloud services need almost no initial investment, because companies simply pay for what they use. On-premise data centers, in contrast, require heavy spending on servers, cooling systems, power backup and buildings before any work begins.

As for recurring overheads, cloud bills can grow quickly if usage is not monitored, whereas an on-premise setup has predictable costs once it is built, although it still needs staff, electricity and maintenance. Therefore, cloud suits changing workloads, while on-premise suits stable and long-term demand.

Workload scalability strongly favours the cloud. Companies can add computing power in minutes during simulations or product launches and reduce it afterwards. On the other hand, on-premise systems have fixed capacity, so expanding them takes weeks or months and leaves hardware idle during quiet periods.

Data security and compliance are more complex. On-premise data centers give the company full physical control, which is important for defence, healthcare or confidential design data. Cloud providers offer strong security tools and certifications, but the company depends on a third party and must follow data-location laws carefully.

In conclusion, the cloud is better for flexibility, speed and low starting cost, while on-premise infrastructure is better for control and predictable long-term expenses. Many enterprises now choose a hybrid model that combines the advantages of both.

Question 4: 3D Printing versus Traditional CNC Milling

3D Printing (Additive Manufacturing) versus CNC Milling (Subtractive Manufacturing) in Prototype Fabrication

In prototype fabrication, 3D printing and CNC milling follow opposite principles. A 3D printer builds a part layer by layer, whereas a CNC machine cuts material away from a solid block. Each method has positives and negatives.

Material wastage is the first point of contrast. 3D printing uses only the material needed for the part, so wastage is low. CNC milling, on the other hand, removes a large amount of material as chips, which increases the cost of expensive metals.

When it comes to production speed, the picture changes with batch size. For single prototypes and small batches, 3D printing is faster because it needs no tooling or setup. For larger batch runs, however, CNC milling is quicker, as it can produce identical parts continuously at high speed.

Surface finish and accuracy favour CNC milling. It produces smooth surfaces and very tight tolerances, so parts often need no extra work. In contrast, 3D printed parts often show layer lines and need sanding or polishing, and their accuracy is lower.

Geometric complexity is the strength of 3D printing. It can easily create hollow parts, internal channels and lattice structures that are impossible to cut with a rotating tool. CNC milling is limited by tool access and the direction of cutting.

In conclusion, 3D printing is ideal for complex, low-volume prototypes with little waste, whereas CNC milling is better for accurate, smooth and larger-batch parts. Engineers often use both methods together to get the best result.

Worksheet 2 – Topic 6: Personal Email Writing

Question 1: Email to a high school mentor/teacher about the first semester

To: [teacher's email address]
Subject: My first semester at engineering college

Dear Sir/Madam,

I hope you are doing well. I wanted to write to you because your guidance in school gave me the confidence to start my engineering journey, and I would like to share how my first semester has been.

The biggest change was the move to independent campus life. Nobody reminds us about deadlines now, so I had to learn to manage my own time, plan my study hours and take responsibility for my work. It was difficult at first, but I am getting used to it.

In our laboratory sessions, we worked on our first small projects, such as basic programs in Python and C and simple circuits. Seeing my own code run correctly gave me a great sense of satisfaction, and I realised how different practical learning is from reading theory.

For the end-semester examinations, I have made a timetable that gives equal time to every subject. I revise the notes every weekend, solve previous question papers and practise the lab experiments so that I do not forget the procedures.

Thank you for everything you taught me. I will be very happy to hear from you and to receive any advice for the coming semesters.

With respect and regards,
B Idhayachandran

Question 2: Email to a high school science teacher about the first term

To: [science teacher's email address]
Subject: Reflections on my first term in engineering

Dear Sir/Madam,

I hope you are keeping well. It has been a few months since I joined engineering college, and I wanted to share my first-term experience with you, since you were the one who made science exciting for me.

The first thing I noticed was how different the hands-on lab coursework is. In school, we mostly saw experiments in a few periods, but here we handle components, tools and software ourselves. I made mistakes in the beginning, but each mistake taught me something and I am becoming more careful and confident.

I also took part in my first technical club activity and a small hackathon. Working with a team to build a solution within a limited time was challenging and thrilling, and I learned a lot about teamwork and debugging from my seniors.

For the upcoming university theory and practical exams, my strategy is simple. I will finish revising each unit at least two weeks before the exams, practise numerical problems daily, and repeat the lab experiments so that I can answer viva questions without fear.

I would be grateful for any advice you may have. Please take care.

Yours sincerely,
B Idhayachandran

Question 3: Email to an elder sibling/alumnus seeking advice

To: [sibling's or alumnus's email address]
Subject: Need your advice on balancing studies and technical clubs

Dear [Name],

How are you? I hope college and work are going well. I am writing because I need your advice on something I am struggling with.

My day starts at 6:30 a.m., and I attend classes until 4:30 p.m. After that, I go to my technical club meetings for about two hours, and in the evening I try to study and finish my assignments. By the time I am free, I am usually very tired.

My biggest challenge is managing the lab record submissions. Because the records must be completed every week, they pile up, and I often finish them at the last minute, which affects the quality.

Since you have handled the same situation, could you please tell me how you balanced your academics with the clubs? I would also like your tips for semester exam revision, such as how to plan the timetable, which subjects to start with, and how to use previous question papers.

I am looking forward to your reply. Thank you in advance, and please take care.

Your loving brother/sister,
Idhaya

Worksheet 2 – Topic 7: Travel Planning and Budgetary Itineraries

Question 1: 4-day tour to an aerospace and satellite launch station

A 4-Day Student Technical Tour to an Aerospace and Satellite Launch Station

Our department plans a four-day tour to a satellite launch station for forty students and two faculty members. We will travel by a chartered train coach and local bus, so that the cost is reasonable.

Budget (per student): train fare ₹1,200, local bus ₹400, accommodation for three nights ₹1,500, food ₹1,200, entry and miscellaneous charges ₹400. The total is ₹4,700, which gives about ₹1,88,000 for the full group of forty.

Day 1: Overnight travel and arrival, check-in at the guest house and an orientation briefing. Day 2: Visit the launch complex viewing gallery, the rocket assembly area and the mission control centre, where scientists explain how a launch is planned and tracked. Day 3: Visit the satellite integration and testing laboratory and the space museum, followed by an interaction with engineers on careers in aerospace. Day 4: Morning group discussion, preparation of our tour report, and the return journey.

Safety protocols: Carry your ID cards and keep them visible at all times. Follow the instructions of the security staff, avoid photography in restricted areas, and stay with the group. Wear closed shoes and drink plenty of water. Keep the emergency contact numbers of the faculty coordinators and follow the launch-site rules strictly.

This tour will give us a first-hand view of how rockets and satellites are built and operated.

Question 2: 4-day tour to a major commercial port and maritime dry-dock facility

A 4-Day Industrial Exposure Tour to a Major Commercial Port and Dry-Dock Facility

Our class of forty first-year students and two faculty members will visit a major commercial port and a dry-dock facility over four days. We will travel in a hired bus, and stay in a budget hotel near the port.

Estimated cost per student: travel ₹1,500, stay for three nights ₹1,800, food ₹1,200, entry passes and miscellaneous ₹500. The total is ₹5,000 per student.

Day 1: Travel to the port city, check in and a short orientation. Day 2: Technical briefing at the port authority, followed by a visit to the container terminal, where cranes, conveyors and tracking systems handle cargo. Day 3: Visit the dry-dock, where ships are lifted and repaired, and see how hull cleaning, welding and engine maintenance are done. An engineer will also brief us on ship stability. Day 4: Visit the control tower and the vessel traffic system, a feedback session and the return journey.

Dockyard safety protocols: Wear helmets, safety shoes and high-visibility jackets at all times. Do not go near moving cranes, cargo or the edge of the dock. Stay in the group, obey the guides and never touch machinery. Keep emergency contacts ready, and carry first-aid supplies.

This tour will help us understand how ports and shipyards work in real life.

Question 3: 4-day tour to an electric locomotive manufacturing works and railway signalling center

A 4-Day Technical Study Tour to an Electric Locomotive Works and a Railway Signalling Center

We plan a four-day study tour for thirty-five students and two faculty members to an electric locomotive manufacturing works and an automated railway signalling center. We will book sleeper-class seats on trains for both journeys, and use a local bus for the visits.

Category-wise expenditure per student: rail tickets ₹1,400, local transport ₹300, accommodation for three nights ₹1,500, food ₹1,200, and entry and miscellaneous ₹300. The total is ₹4,700.

Day 1: Overnight train journey and arrival. Day 2: Visit the locomotive works, where we will see the shop-floor stages such as shell fabrication, wiring, traction motor assembly and final testing of the engine. Day 3: Visit the automated signalling center, where engineers explain how sensors, track circuits and computers control train movement and keep the trains safe. Day 4: Interaction with engineers, preparation of a tour report and the return journey.

Site safety rules: Wear helmets, safety shoes and closed clothing in the shop floor areas. Never walk on the railway tracks, and keep a safe distance from cranes and live electrical equipment. Follow the guides' directions and stay with the group. Do not touch any machine or signal equipment, and report any problem immediately to the faculty coordinator.

This tour will connect our classroom learning with real railway engineering.

Experiment No: II-7

Project Proposal: Ideation

LunArk: Genesis, Technical Specification

Team LunArk | NASA Space Apps Challenge 2026 | Draft v1, October 17, 2026

Status: draft. Items marked [default] are working assumptions the team can change. Items marked [open] still need a decision.

1. Overview

LunArk: Genesis is a desktop game where the player runs a lunar outpost. You place buildings, keep power flowing through the long lunar night, protect the base from radiation, earn money and stop the crew from getting bored. The aim is to let students aged 13-16 feel the engineering trade-offs that decide whether a real Moon mission succeeds or fails.

The game is built in Python with pygame-ce. All art is hand-drawn pixel art made in Aseprite. The code is written with help from an AI coding agent (Google Antigravity) and reviewed by the team, and the project submission discloses this.

2. Goals and non-goals

2.1 Goals

- Make real mission trade-offs tangible: power, life support, radiation shielding, money and crew morale.
- A 13-16 year old should understand it within about ten minutes, thanks to a guided tutorial.
- It runs on Windows, macOS and Linux with two commands and no installer.
- It is fully documented, so a judge or a new teammate can run, read and extend it.
- Game logic, user interface and game data stay strictly separate.

2.2 Non-goals

- No multiplayer, online features or accounts.
- No packaged executables (no PyInstaller). Players run the game from source.
- No AI-generated art, fonts or audio. The team makes all art.
- No zooming or rotating of sprites, which keeps the pixel art crisp.
- No real orbital mechanics. The science content is simplified, but it must be accurate.

3. Target audience and design feel

- Audience: students aged 13-16, with little or no engineering background.
- Feel [default]: calm and hopeful at the start, with pressure rising later, but never unfair. When players fail, they should think “I see what I did wrong”, not “that was random”. Failure screens always show the exact reason and a hint.
- Inspiration: the mobile game Spaceflight Simulator, used only for ideas. Those are a clean, flat, icon-first interface; building from parts with a ghost preview; a side-view 2D world; an always-visible time warp that slows itself down when something important happens; and predictive information such as countdowns and warnings. No art, code, name or exact UI is copied.

4. Technology stack

Item	Decision
Language	Python 3 (minimum tested version to be recorded in the README)
Library	pygame-ce, version pinned in requirements.txt
Testing	pytest (development only, in requirements-dev.txt)
Art tool	Aseprite, exported as PNG
Packaging	None. Run from source
Platforms	Windows, macOS, Linux, desktop only

4.1 How players run the game

```
pip install -r requirements.txt
python3 main.py
```

On Windows the second command may be python main.py or py main.py. The README documents all three, along with optional virtual-environment steps.

5. Repository structure

```
LunArkGenesis/
  main.py                entry point
  requirements.txt       pygame-ce only, pinned
  requirements-dev.txt   pytest
  README.md              full documentation
  LICENSE                GPL-3.0
  fonts/                 pixel font (team-supplied)
  img/                   all art (team-made)
    logo/ ui/ icons/ terrain/ sky/ buildings/
    characters/ cutscene/ events/ fx/ vehicles/
  saves/                 runtime only, gitignored
  tests/                 pytest tests for game/logic
  game/
    app.py               window, main loop, screen manager
    constants.py         fixed numbers, colors, paths, index tables
    assets.py            image/font loading, placeholders, 9-slice
    logic/               game rules, no pygame imports
    ui/                  drawing and input
    screens/             splash, title, settings, cutscene, play, pause, end
    data/                fixed game content as JSON
```

Rule of thumb: logic decides what happens, ui shows it, data holds the numbers and text, and saves holds the player’s files.

6. Architecture

6.1 Layers

Layer	Responsibility	Rules
game/logic	Bars, clock, world grid, buildings, events, win/lose, settings, save/load	Never imports pygame. Fully unit-tested
game/ui	Screens, widgets, drawing, input, display scaling	Contains no game rules
game/data	buildings.json, events.json, balance.json, story.json	Numbers and text only
game/assets.py	Loads images and fonts	Draws a labelled placeholder when a file is missing
saves/	settings.json, savegame.json	Created at runtime, validated on load

6.2 Main loop

The app owns one window and a screen manager. Each screen implements handle events, update and draw. The simulation moves forward in fixed logic ticks (1 tick = 1 second at 1x speed), independent of the frame rate, so the FPS setting never changes game speed.

6.3 Error handling

A missing or corrupt save, settings file, JSON file or image must never crash the game. It falls back to defaults or placeholders and logs a clear message. At startup it prints a single summary of missing assets.

7. Display, fullscreen and settings

7.1 Rendering

- The game draws to an internal canvas of 480 x 270 [default] (16:9) with 16 x 16 pixel tiles.
- The canvas is scaled to the window or screen with nearest-neighbor scaling. The aspect ratio is kept, and black bars fill any gaps.
- Mouse positions are converted back to canvas coordinates.
- An optional integer scaling setting uses whole-number scale factors only, which gives sharper pixels but larger bars. Default: on.

7.2 Display modes

- The first launch starts in fullscreen at the desktop resolution, and the choice is remembered.
- F11 toggles between fullscreen and windowed mode from any screen.
- Esc never quits instantly. It goes back in menus and opens the pause menu in the game. Quit is a button on the title and pause menus.
- On Windows the process is made DPI-aware so sizes are correct on scaled displays.

7.3 Settings menu

Available from the title screen and the pause menu, by mouse or keyboard.

Setting	Options
Display mode	Fullscreen, Windowed
Resolution (windowed size)	1280x720, 1366x768, 1600x900, 1920x1080, 2560x1440, Native desktop. Only sizes that fit the monitor are listed. Greyed out in fullscreen
FPS cap	30, 60, 120, 144, Unlimited. Default 60
VSync	On, Off. Default on. Hidden if unsupported
Show FPS counter	On, Off. Default off
Integer scaling	On, Off. Default on
Buttons	Apply, Back, Reset to defaults

Changing the display mode or resolution brings up a “Keep these settings? (reverting in 10s)” prompt that reverts on its own, so nobody gets stuck on a bad screen. Settings are saved to saves/settings.json, validated on load, and replaced by defaults if invalid.

8. Screen flow

1. Splash: a black screen. The logo (33 x 33 px, integer-scaled with nearest-neighbor) fades in and out over about 3 seconds. Any key or click skips it.
2. Title screen: wordmark and background. Menu: New Game, Continue (disabled without a save), Settings, Quit.
3. Intro cutscene: a full-screen picture and a dialogue box with typewriter text. The player wakes up to a man looking at them. He welcomes them to the Moon and asks their name (typed, maximum 12 characters, default “Commander”). He explains that NASA sent them to establish a base and introduces himself as Lieutenant Hale [default name], then walks through the first steps. The cutscene can be skipped.
4. Guided tutorial: inside the real game. The lieutenant waits for the player to finish each step: place a Fuel Cell Generator, place the Habitat Pod, watch the Energy bar, pan the camera, build a Lab, understand Boredom and build a Gym, understand the Event bar, read the sunset countdown, then dig a building underground and see its radiation exposure improve.
5. Main game.
6. Pause menu: Resume, Settings, Save, Quit to Title, Quit Game.
7. End screens: win or lose, always with the exact reason and a hint for next time. Buttons: Retry, Title.

9. Game design

9.1 The world

- A side-view cross-section of the Moon, 48 columns wide [default], scrolling horizontally on the 480 px canvas.
- From top to bottom: black sky (the lunar sky is black even in daytime), the ground line, then 4 rows [default] of underground regolith.
- The sky shows stars, the Sun crossing during the day, and Earth hanging almost still. Two parallax horizon layers add depth.
- The camera pans horizontally only (A and D, arrow keys, or dragging with the middle mouse button). There is no zoom.

9.2 Placing buildings

- The player picks a building from a collapsible build tray at the bottom of the screen.

- A ghost preview follows the mouse, snapped to the grid. It turns green when the spot is valid and red when it isn't, with a one-line reason such as "Needs the surface", "Not enough cash", "Locked: needs Science 50" or "Space is taken".
- Left click places, and right click or Esc cancels. Demolishing refunds part of the cost.
- Each building has a placement type: surface, underground or either.
- Building underground digs the footprint automatically. The cost is the building cost plus a dig cost per tile. No access shafts are modeled, which is a deliberate abstraction stated in the README.
- Radiation exposure runs from 1.0 on the surface down to about 0.05 when deep, depending on how many regolith rows sit above the building. A small radiation icon shows it.
- Selecting a building opens an info card with its role, energy in and out, exposure, status, a science fact, and Repair or Demolish buttons.

9.3 Time and day-night cycle

- 1 tick equals 1 second at 1x speed. Time warp buttons: Pause, 1x, 2x, 4x, 8x.
- Warp drops back to 1x automatically, with an on-screen reason, when a warning or event occurs or when Energy falls below 15% [default].
- The real lunar night lasts about 14 Earth days, so it is compressed: 180 ticks of sunlight, then 180 ticks of night [default], set in balance.json.
- The HUD shows a sun or moon icon and a countdown to the next sunset or sunrise.

9.4 The five bars

Bar	Meaning
Energy	Stored energy between 0 and a capacity. Each tick: production minus consumption. At 0, buildings shut down in priority order, with the Habitat Pod last
Science	Accumulated research points. Used for the win goal and for unlocking equipment tiers
Boredom	0 to 100. Rises over time with crew size. Activity buildings lower it
Revenue	Cash balance. Spent on building, digging and repairs, earned by Cashflow buildings. Shown as a bar toward the next unlock plus a number
Event	0 to 100. Drifts up when Boredom is high and down when Boredom is low. A higher value means a higher chance per tick of a random failure event

Bars flash when they run low or turn dangerous. All rates and thresholds live in balance.json, so they can be tuned without code changes.

9.5 Buildings

There are four categories with three tiers each. Tiers 2 and 3 unlock only when both Science and Revenue pass thresholds set in balance.json.

Category	Tier 1	Tier 2	Tier 3	Rule
Generator	Fuel Cell Generator (2x2)	Solar Array (3x2)	Nuclear Plant (3x3)	Produces Energy
Cashflow	Datacentre (2x2)	Precious Metal Mining (3x2)	Helium-3 Extraction (3x3)	Uses Energy, produces cash
Activity	Gym (2x2)	Cafe (2x2)	Restaurant (3x2)	Uses Energy, lowers Boredom
Science	Lab (2x2)	Land Research (3x2)	Planet Stats (3x3)	Uses Energy, produces Science

Special building: the Habitat Pod (3x2) provides life support and housing. It must stay powered, and it is placed during the tutorial.

Default placement: surface for Fuel Cell Generator, Solar Array, Precious Metal Mining, Helium-3 Extraction, Land Research and Planet Stats. Either for Habitat Pod, Datacentre, Gym, Cafe, Restaurant, Lab and Nuclear Plant.

Science accuracy rules:

- The first generator is a fuel cell, not a diesel generator. Burning fuel needs oxygen, and the Moon has no air. The fuel is shipped from Earth, so it carries a running cash cost per tick.

- The Solar Array produces nothing at night, and dust storms reduce its output. That is the central trade-off against the expensive, constant Nuclear Plant.
- The Nuclear Plant has high constant output and a very high cost, and it is locked behind Science.
- The lunar resource is Helium-3, not Helium II.
- Every building and event carries a short science fact written for ages 13-16, flagged verified: false until the team checks it against NASA sources.

9.6 Events

Each event has an id, name, description, effect, minimum Event bar level, warning time, preparation tip and science fact.

Event	Effect
Solar Flare	Warning about 20 ticks ahead. Each building rolls a chance of going offline equal to a base chance times its radiation exposure. A surface Habitat Pod makes the crew shelter, pausing Science and spiking Boredom. Buried buildings are mostly safe
Dust Storm	Solar Array output halved for some ticks
Equipment Failure	One random building goes offline until repaired
Micrometeorite	Small cash loss
Crew Argument	Boredom spike

Warnings show a banner with a countdown. Events show a popup that says what happened, why, and a tip. Repairs cost cash and are done from the info card.

9.7 Win and lose conditions [default]

- Win: build Planet Stats and keep it running for a short period.
- Lose, life support: the Habitat Pod stays unpowered for more than a set number of ticks.
- Lose, stranded: zero Energy, zero cash and no way to earn either.

9.8 HUD layout

Fitted to 480 x 270. The five bars with icons run along the top left. Time warp and the sun or moon countdown sit at the top right, next to a settings gear. A collapsible build tray at the bottom center has five category tabs (Generator, Cashflow, Activity, Science, Habitat). The info card appears on the right when something is selected, and a small message log sits at the bottom left. Icons come first and text second, with tooltips on hover.

10. Data files

All game content lives in game/data as JSON, so it can be tuned without touching code.

File	Contents
buildings.json	id, name, category, tier, cost, placement, footprint, energy use or production, cash per tick, science per tick, boredom reduction, sprite, description, science fact, verified flag
events.json	id, name, description, effect, minimum Event bar, warning ticks, preparation tip, science fact, verified flag
balance.json	day and night lengths, bar rates, unlock thresholds, dig cost per tile, refund rate, radiation exposure table, warp auto-slow threshold, win and lose thresholds
story.json	all cutscene and tutorial lines, mentor name, placeholders for the player’s name

Building ids, which are also the sprite file names: habitat_pod, fuel_cell_generator, solar_array, nuclear_plant, datacentre, precious_metal_mining, helium3_extraction, gym, cafe, restaurant, lab, land_research, planet_stats.

11. Art pipeline

All art is hand-drawn in Aseprite and exported as PNG. The loader accepts any file placed at the documented path, with no code change. Animated sprites are horizontal strips of equal-size frames, and the frame count is detected automatically from the image width. For buildings, frame 0 is offline and later frames are the running animation. Missing art shows a labelled placeholder of the correct size.

Area	Path and size
Logo	img/logo/logo.png, 33x33
UI pieces	img/ui/: 9-slice panel (24x24), button (4 states, 96x24), bar frame, dialogue box, toggle, next arrow, cursor, title, title background, end screens
Icons	img/icons/icons.png, one 128x48 sheet of 24 icons at 16x16 in a fixed order
Terrain	img/terrain/tileset.png, 128x48, 24 tiles at 16x16
Sky	img/sky/: stars, earth, sun, far and near horizon
Buildings	img/buildings/<id>.png, frame size equals footprint times 16 px
Characters	img/characters/mentor.png (4 expressions at 64x64), astronaut (optional)
Cutscene and events	img/cutscene/ at 480x270, img/events/ at 96x64
Effects and vehicles	img/fx/, img/vehicles/ (optional)
Font	fonts/*.ttf, a pixel font whose license has been checked and credited

The separate Asset Guide describes every sprite, its size, its priority (must, should, nice) and a suggested drawing order. About 15 files make a playable demo.

12. Testing and quality

- Every rule in game/logic has a pytest test, and tests must pass before a phase is finished.
- A terminal runner simulates a game as text, so the balance can be tuned before any graphics exist.
- Each phase has a manual test checklist: fullscreen, F11, resolution change, the auto-revert confirmation, and running with no art files at all.
- An F3 debug overlay lists which assets are still placeholders.

13. Documentation requirements

The README is written and updated after every phase. It contains a description and screenshot, a 30-second quick start, controls and settings, how to play, the science behind the game, a plain-language tour of every folder and file, the architecture, how-to guides (add a building, event, balance change, art, font or screen), data schemas, how to run tests, the art guide, the AI disclosure, known issues, roadmap, changelog, credits and license.

All code has type hints and docstrings, and comments explain why, not what.

14. Use of AI and disclosure

- Code is written with the AI coding agent Google Antigravity, one phase at a time, and the team reviews it before the next phase starts.
- Every Python file carries a header comment saying it is AI-assisted.
- All art is made by hand in Aseprite. The agent is forbidden from creating image, font or audio files.
- The README and the Space Apps submission form say where AI was used, as the event requires.
- The team must be able to explain every file to the judges.

15. Development plan

Phase	Content	Target
1	Scaffold, display system, settings menu, splash, title screen, README v1	by Oct 22
2	Headless game logic, data files, tests, terminal runner, balance tuning	by Oct 28
3	Side-view play screen: world, camera, build tray, ghost preview, digging, HUD, time warp, info card, save and load	by Nov 4
4	Intro cutscene, name input, guided tutorial	by Nov 8
5	Warnings, event popups, forecast, end screens, polish, debug overlay, QA, final README	by Nov 11
Freeze	No new features. Demo video, final README, submission	Nov 12-13
Event	Space Apps weekend	Nov 14-15
6 (stretch)	Resupply lander, minimap, battery building, sound hooks, extra events	only if time allows

Art is drawn in parallel, starting with the minimum set. If time runs short, the full-scale game is cut back from Phase 5 and Phase 6 first.

16. Risks and mitigations

Risk	Mitigation
Event rules may not allow work done before the event [open]	Ask the local organizer. If pre-building is not allowed, use the time for planning, art and learning, and build during the 48 hours
Scope is too large for the time	Phased plan, strict freeze date, cut stretch items first
The AI agent produces code the team does not understand	One phase at a time, read every diff, ask the agent to explain files
Balance feels wrong	Data-driven numbers and a text-only simulation before the UI
Art is not ready in time	Placeholders keep the game playable; draw the minimum set first
Judges cannot run the game	Clear quick start, demo video, tested on a clean machine
Fullscreen gets stuck on someone’s machine	F11 and Esc always work, and display changes auto-revert
Science facts are wrong	Mark all facts unverified until checked against NASA sources
AI usage limits are reached mid-build	Keep a backup agent or local model ready, commit after every working step

17. Open questions

1. Is work done before the event allowed? [open, confirm with the organizer]
2. Are time and tile sizes final (480 x 270 canvas, 16 x 16 tiles)?
3. Is the game real-time with time warp (the current default) or turn-based?
4. What is the mentor’s final name?
5. What are the final win and lose rules and balance numbers?
6. Which pixel font will be used, and under what license?

18. Glossary

- Regolith: the loose dust and rock covering the Moon’s surface.
- Tick: one step of the game simulation, equal to one second at 1x speed.
- 9-slice: a way to stretch a UI image without distorting its corners.
- Exposure: how much radiation reaches a building, from 1.0 (surface) down to about 0.05 (deeply buried).
- Ghost preview: the see-through building that follows the mouse before it is placed.
- Time warp: speeding up the simulation clock.

Experiment No: II-8

LinkedIn and GitHub Profiles

Part A: LinkedIn profile

B Idhayachandran

Aspiring Cybersecurity Analyst and Data Scientist

CSE Student (Batch 2026·2030) | Python & C | Cybersecurity | Data Science | github.com/idhayachandran

About

I'm a curious Computer Science Engineering student who enjoys finding out how systems work and how they fail. Most of my work so far is in Python and C, and I like to think a problem all the way through before I build anything.

I'm interested in cybersecurity and data science because they fit together: security teams collect huge amounts of data, and data science helps turn it into evidence. My long-term goal is to work on threat detection, using analytics and machine learning to protect systems and people's information.

Focus areas: Cybersecurity • Data science • Python • C • Linux • Networking fundamentals • Statistics

Featured Projects

[Project 1 name]
[One or two lines on what it does and what you learned.]
Tech: Python | github.com/idhayachandran/[repo]

[Project 2 name]
[One or two lines on what it does and what you learned.]
Tech: [add] | github.com/idhayachandran/[repo]

Education

B.E. Computer Science & Engineering (CSE), 1st Year, St. Joseph's College of Engineering, Chennai. Batch 2026 – 2030.

Honors & Awards

[Add any awards or competitions here]

Skills

Languages	Python, C
Operating Systems	Linux, Windows
Tools	Git, GitHub, Jupyter Notebook (learning)
Strengths	Problem solving, curiosity, self-directed learning

Licenses & Certifications (Planned)

- Entry-level cybersecurity certificate (for example Google Cybersecurity or ISC2 Certified in Cybersecurity) – target: before Year 2
- CompTIA Security+ or similar – target: Year 2
- Cloud security or data analyst certificate – target: Year 3

Career Roadmap

Year	Focus	Deliverable
Year 1 (2026–27)	Linux, networking, Python, SQL basics	GitHub portfolio with a log-analysis notebook
Year 2 (2027–28)	Statistics, machine learning, web and network security	Anomaly-detection project on a public dataset
Year 3 (2028–29)	Threat detection, SIEM tools, cloud security	Mini security-monitoring dashboard
Year 4 (2029–30)	Security analytics, leadership	Capstone: ML-based threat detection

Connect

GitHub: github.com/idhayachandran | LinkedIn: linkedin.com/in/idhayachandran-b-27b443440

Part B: GitHub profile README

Hi there, I’m Idhaya (B Idhayachandran)

CSE Student | Cybersecurity & Data Science Enthusiast | Python & C

LinkedIn: Idhayachandran | GitHub: idhayachandran

About Me

- I’m a Computer Science Engineering student, building a foundation in programming, systems and data.
- I’m working toward a career in cybersecurity, using data science to detect and prevent threats.
- I like understanding how systems work and fail, and I try to document what I learn.
- Currently learning: Linux, networking basics, Python for data analysis and SQL.

Tech Stack & Tooling

Domain	Technologies & Tools
Languages	Python, C
Operating Systems	Linux, Windows
Data	Python, Jupyter Notebook (learning pandas, NumPy, SQL)
Version Control	Git, GitHub

Featured Repositories

[Repository 1] ([link])
[Short description]
Python

[Repository 2] ([link])
[Short description]
[Tech]

Career Roadmap

- Done: started building programming fundamentals in Python and C.
- In progress: Linux, networking, basic statistics and data analysis.
- Goal: earn an entry-level cybersecurity or data certificate before Year 2.
- Goal: build a security-focused data project, then grow toward threat detection and analytics.

Certifications & Achievements

[Add when earned]

Connect With Me

I’m happy to connect with fellow students and engineers interested in cybersecurity and data science. Reach out on LinkedIn: Idhayachandran